

Z Binding Calculus

txt2tex example

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$[TrackId, AlbumId, ArtistId]$

Track

$trackId : TrackId$
 $albumId : AlbumId$
 $duration : \mathbb{N}$

Album

$albumId : AlbumId$
 $artistId : ArtistId$
 $year : \mathbb{N}$
 $tracks : \mathbb{N}$

Atomic Binding

$\langle\!\langle trackId == t.trackId \rangle\!\rangle$

Binding in Set Comprehension

$\{ t : Track \mid t.duration < 180 \bullet \langle\!\langle trackId == t.trackId \rangle\!\rangle \}$

Multi-Component Binding

$\langle\!\langle trackId == t.trackId, year == a.year, tracks == a.tracks \rangle\!\rangle$

Multi-Variable Comprehension with Binding

$\{ t : Track; a : Album \mid t.albumId = a.albumId \bullet \langle\!\langle trackId == t.trackId, year == a.year, tracks == a.tracks \rangle\!\rangle \}$

Empty Binding

$\langle\!\langle \rangle\!\rangle$